

UNIT-2

RESISTANCE, INDUCTANCE & CAPACITANCE MEASUREMENT



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Wheatstone Bridge

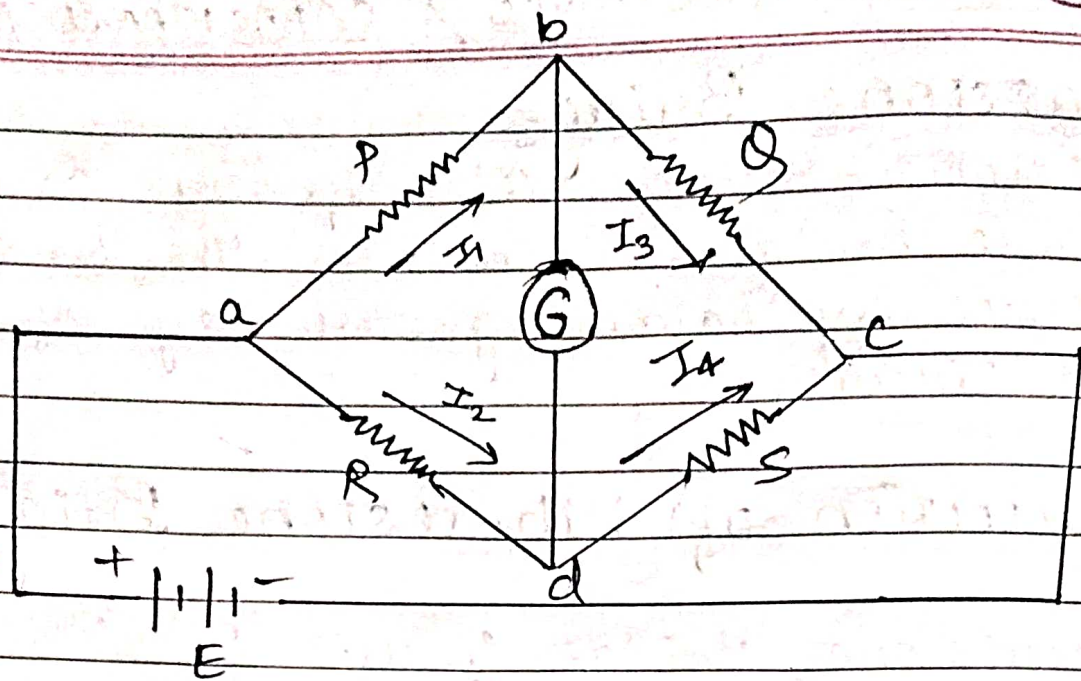
Wheatstone Bridge, also known as the Resistance bridge, calculates the unknown resistance by balancing two legs of the Bridge circuit.

Construction of Wheatstone Bridge

A Wheatstone Bridge circuit consists of four arms, of which two arms consist of known resistance while the other two arms consist of an unknown resistance and a variable resistance. The circuit also consists of a galvanometer and an electromotive force source.

The emf source is attached between points a and b while the galvanometer is connected between points c and d. The current that flows through the galvanometer depends on its potential difference.

Principle - The Wheatstone Bridge works on the principle of null deflection, i.e. the ratio of their resistance is equal, and no current flows through the circuit. Under normal conditions, the bridge is an unbalanced condition where current flows through the galvanometer. The bridge is said to be balanced when no current flows through the galvanometer.



Wheatstone Bridge Derivation -

$$I_1 P = I_2 R \quad \text{--- (1)}$$

$$I_1 = I_3 \Rightarrow \frac{E}{P+Q}$$

$$I_2 \Rightarrow I_4 = \frac{E}{R+S}$$

Here, E is the emf of the battery.

By substituting the value of I_1 & I_2 in eq (1)

$$\frac{PE}{P+Q} \Rightarrow \frac{RE}{R+S}$$

$$\frac{P}{P+Q} \Rightarrow \frac{R}{R+S}$$

$$P(R+S) \Rightarrow R(P+Q)$$

$$PR + PS = RP + RQ$$

$$PS = RQ$$

$$\frac{P}{R} = \frac{Q}{S} \quad \text{--- (2) ---} \rightarrow \text{Balanced Condition}$$

$$R = \frac{P \times S}{Q} \quad \text{--- (3) ---}$$

Value of Unknown Resistance

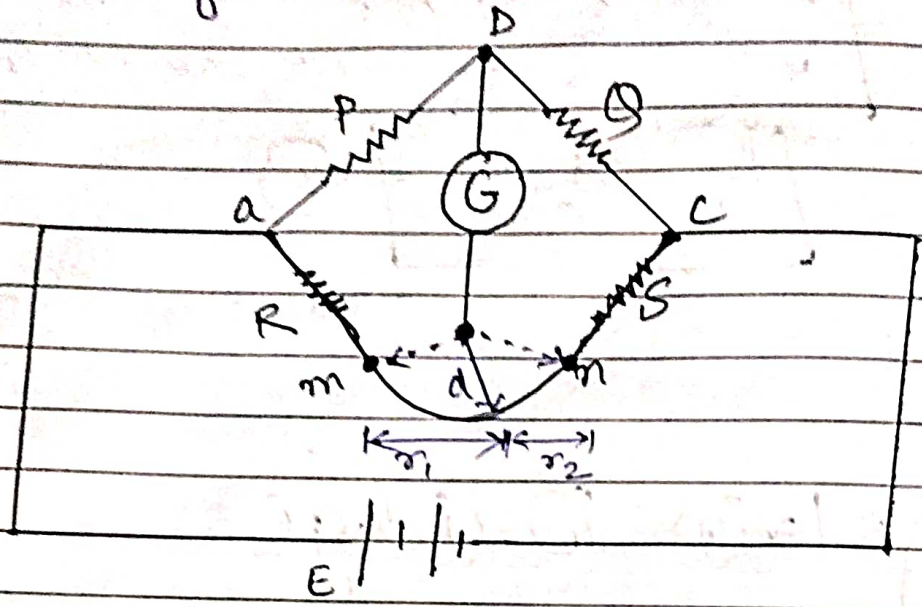
Kelvin Bridge

The Kelvin Bridge or Thompson Bridge is used for measuring the Unknown Resistance having a value less than 1Ω . It is the modified version of the Wheatstone Bridge.

Use - Since we know, Wheatstone bridge use for measuring the resistance from a few ohms to several kilo-ohm. "But error occurs in the result when it is used for measuring the low resistance." This is the reason because of which the Wheatstone bridge is modified, and the Kelvin bridge obtains. "The Kelvin bridge is suitable for measuring the low resistance."

Modification of Wheatstone Bridge

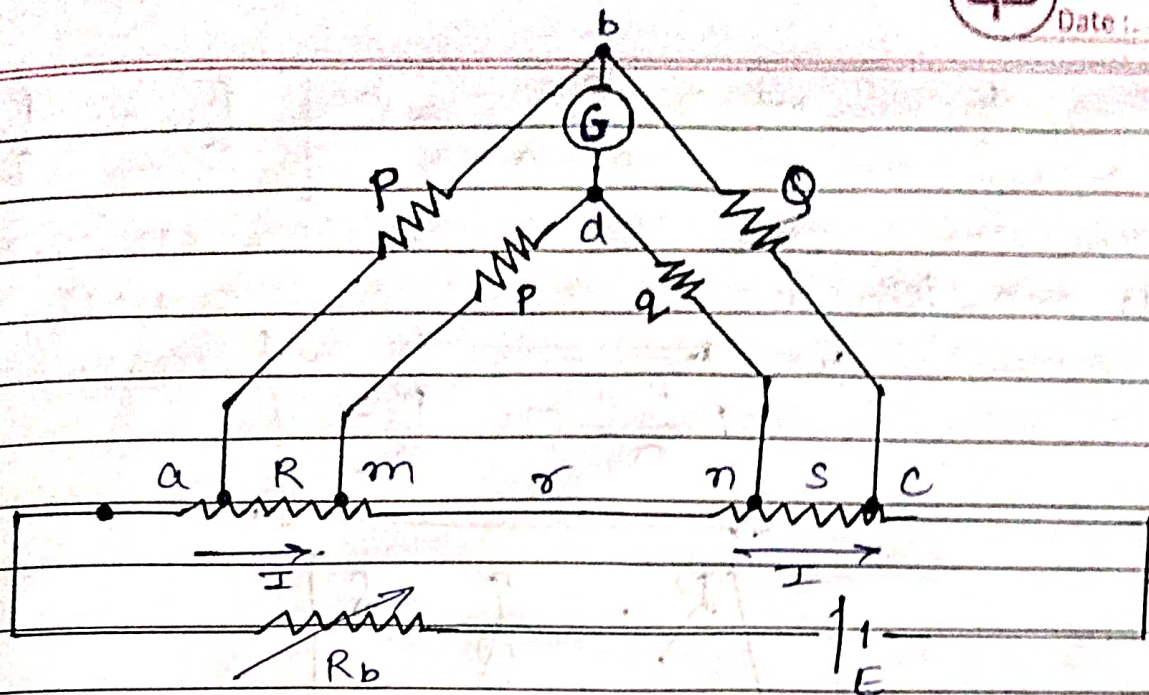
In wheatstone Bridge, while measuring the low-value resistance, the resistance of their lead and contacts increases the resistance of their total measured value.



$$R = \frac{P \cdot S}{Q}$$

Kelvin Double Bridge Circuit

The ratio of the arms P and Q are used to connect the galvanometer at the right place between the point j and K. The j and K reduce the effect of connecting lead. The P and Q is the first ratio of the arm and P and Q is the second arm ratio."



Kelvin Double Bridge

$$E_{ab} \Rightarrow \frac{P}{P+Q} E_{ac}$$

$$E_{ac} \Rightarrow I \left[R + S + \frac{(P+Q)r}{P+Q+r} \right] \quad \text{--- (1)}$$

$$E_{amd} = I \left[R + \frac{P}{(P+Q)} \left\{ \frac{(P+Q)r}{P+Q+r} \right\} \right]$$

$$E_{ad} \Rightarrow I \left[\frac{Pr}{P+Q+r} \right] \quad \text{--- (2)}$$

for zero Galvanometer Deflection -

$$E_{ac} \Rightarrow E_{amd}$$

$$\frac{P}{P+Q} \cdot I \left[R + \frac{P}{(P+Q)} \left\{ \frac{(P+Q)r}{P+Q+r} \right\} \right] = I \left[\frac{Pr}{P+Q+r} \right]$$

$$R = \frac{P \cdot S}{Q} + \frac{P \cdot r}{P + Q + r} \left[\frac{P}{Q} - \frac{P}{Q} \right]$$

As we know;

$$\left[\frac{P}{Q} = \frac{P \cdot r}{P + Q + r} \right]$$

$$\left[R = \frac{P \cdot S}{Q} \right]$$

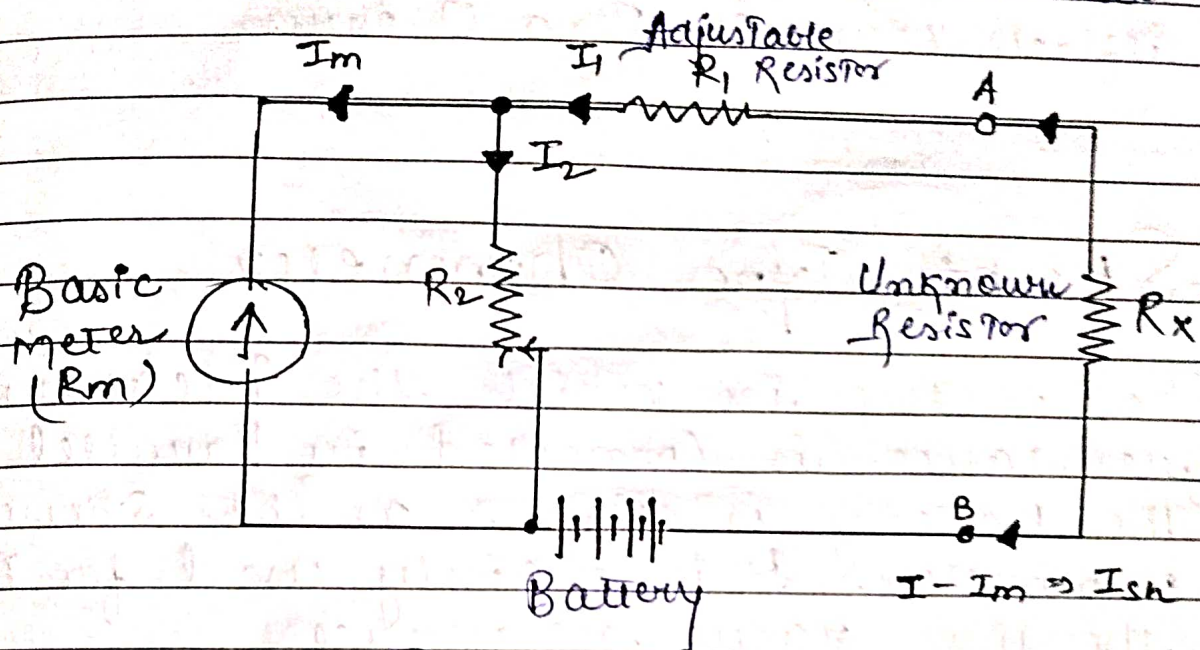
Ohmmeter - The meter which measures the resistance and the continuity of the electrical circuit and their components such type of meter is known as the ohmmeter.

It measures the resistance in ohms. The micro-ohmmeter is used for measuring the low resistance and the mega ohmmeter measures the high resistance of the circuit. The ohmmeter is very convenient to use but less accurate.

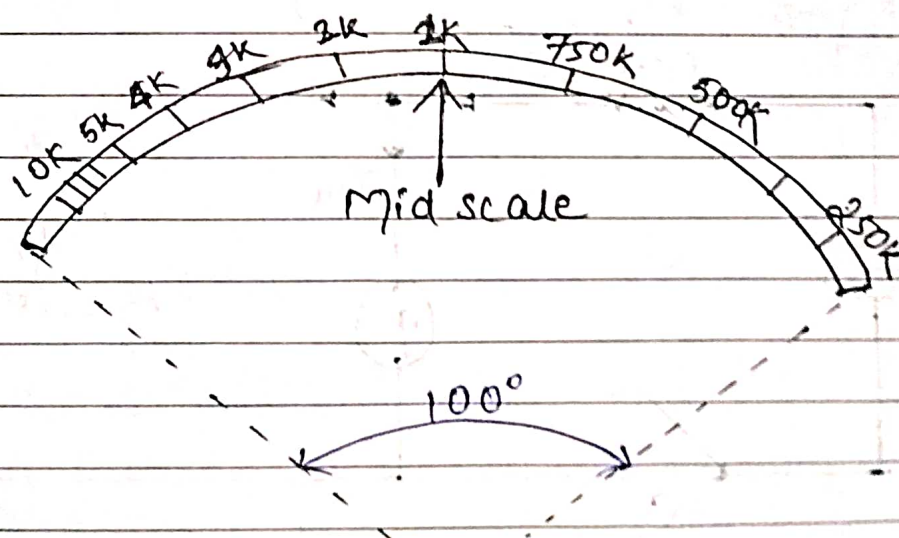
1. Series Ohmmeter - In series ohmmeter, the measuring resistance components or circuit is connected in series with the meter. The value of resistance is measured through the d'Arsonval Movement connected in parallel with the shunt



Resistor R_2 . The parallel resistance R_2 is connected in series with the resistance R_1 and the battery. The component whose resistance is used to be measured is connected in series with the terminal A and B.



Series Type ohmmeter



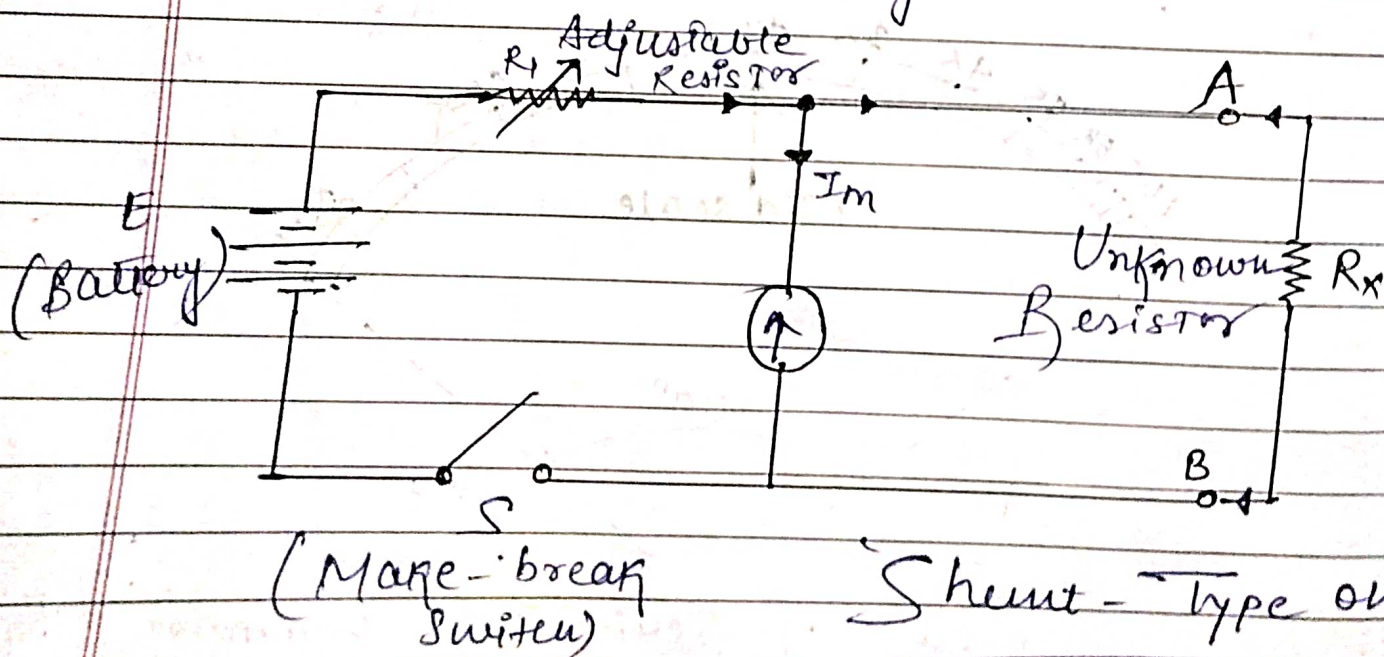
Scale of Series Type ohmmeter.

The value of unknown resistance is zero. Large current flow through the meter. In this condition, the shunt resistance is adjusted until the meter indicates the full load current. For full load current, the pointer deflects towards zero Ω ohms.

Shunt Type Ohmmeter

The meter in which the measuring resistance is connected in parallel with the battery is known as the shunt ohmmeter. It is mainly used for measuring the low-value resistance.

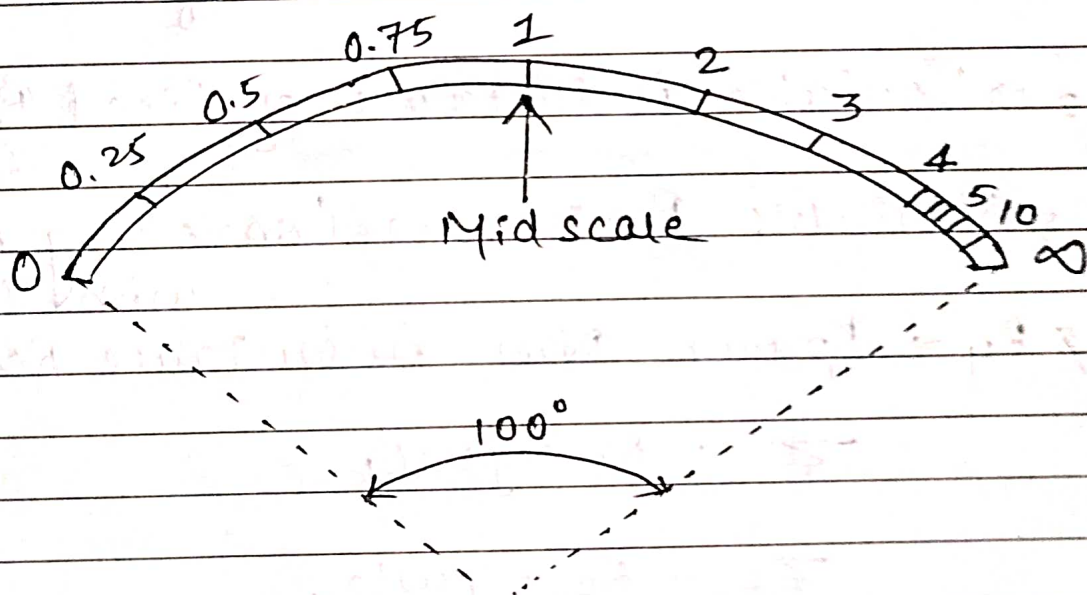
The circuit diagram of the shunt ohmmeter is shown in the figure below -



Shunt - Type ohmmeter

The battery (E), basic meter (R_m) and the adjustable resistance are the main components of the shunt ohmmeter. The unknown resistance is connected across Terminal A and B.

When the value of Unknown Resistance is zero the meter current becomes zero. And if the resistance become infinite (i.e. the Terminal A and B are open) then the current passes through the battery and the Pointer shows the full-scale deflection toward left. The shunt type ohmmeter has the zero mark (No current) on the left of the scale and the infinity mark on their Right Side.



Scale of Shunt-Type ohmmeter

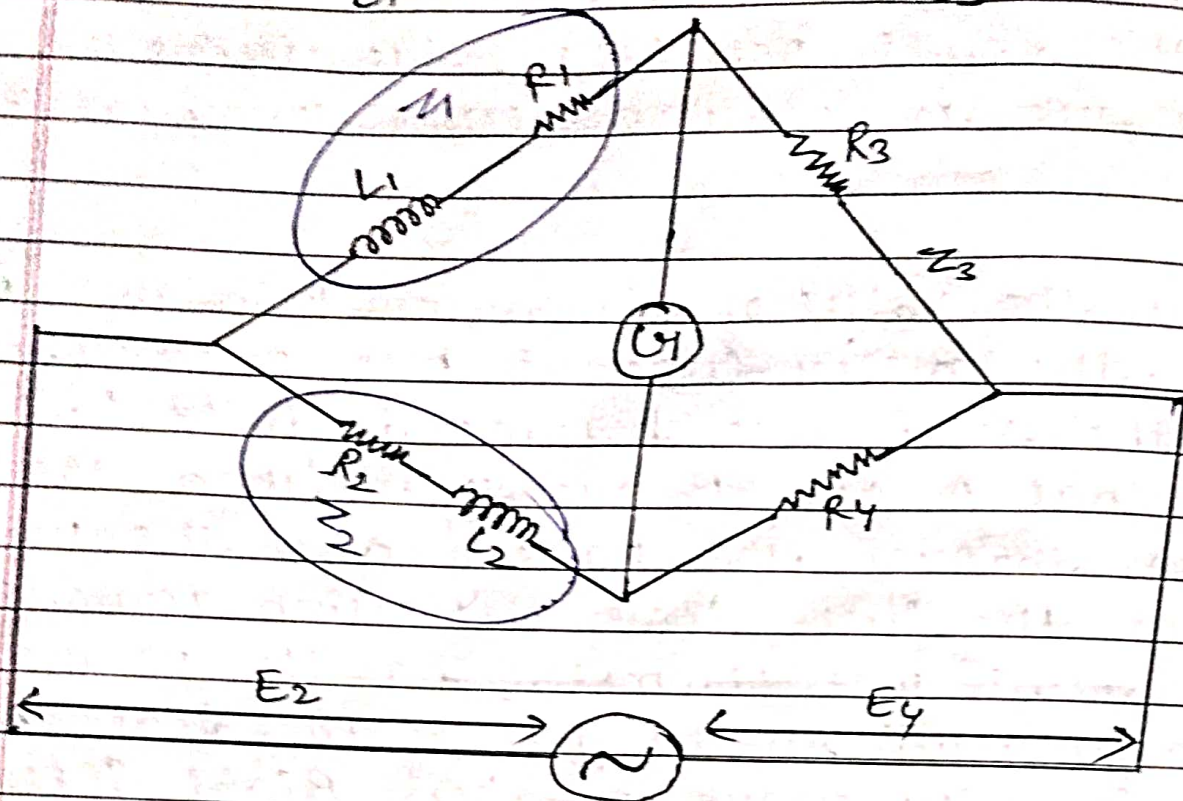
AC Bridge's



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1) Maxwell's Bridge $\rightarrow$ Self Inductance



$L_1 \Rightarrow$ Unknown Inductance of R_1

$L_2 \Rightarrow$ Variable inductance of fixed Resistance R_2

$R_2 \Rightarrow$ Variable Resistance Connecting in series with Inductor L_2

$R_3 \& R_4 \Rightarrow$ Known Non-inductance Resistance.

$$Z_1 \Rightarrow R_1 + j\omega L_1$$

$$Z_2 \Rightarrow R_2 + j\omega L_2$$

$$Z_3 \Rightarrow R_3$$

$$Z_4 \Rightarrow R_4$$

$$Z_1 Z_4 \Rightarrow Z_2 Z_3$$

$$\Rightarrow (R_1 + j\omega L_1) \times (R_4) \Rightarrow (R_2 + j\omega L_2) \times R_3$$

$$\Rightarrow R_1 R_4 + j\omega L_1 R_4 \Rightarrow R_2 R_3 + j\omega L_2 R_3$$

$$\Rightarrow R_1 R_4 - R_2 R_3 \Rightarrow j\omega L_2 R_3 - j\omega L_1 R_4$$

$$\Rightarrow R_1 R_4 - R_2 R_3 \Rightarrow j\omega (L_2 R_3 - L_1 R_4)$$

$$\Rightarrow R_1 R_4 + j\omega L_1 R_4 \Rightarrow j\omega L_2 R_3 + R_2 R_3$$

Comparing the Real Value & Imaginary Value-

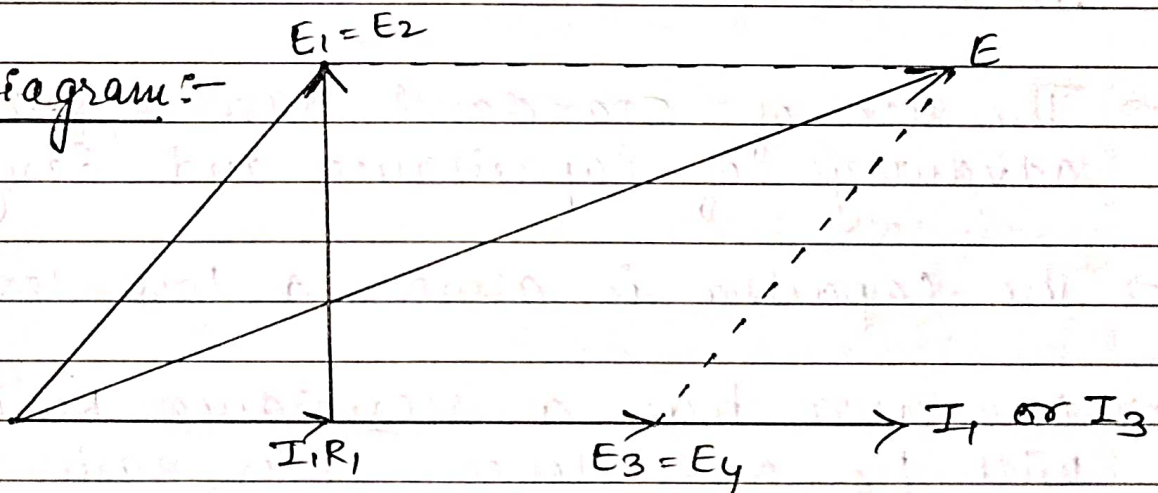
$$\boxed{R_1 R_4 = R_2 R_3}$$

$$\cancel{j\omega L_1 R_4} = \cancel{j\omega L_2 R_3}$$

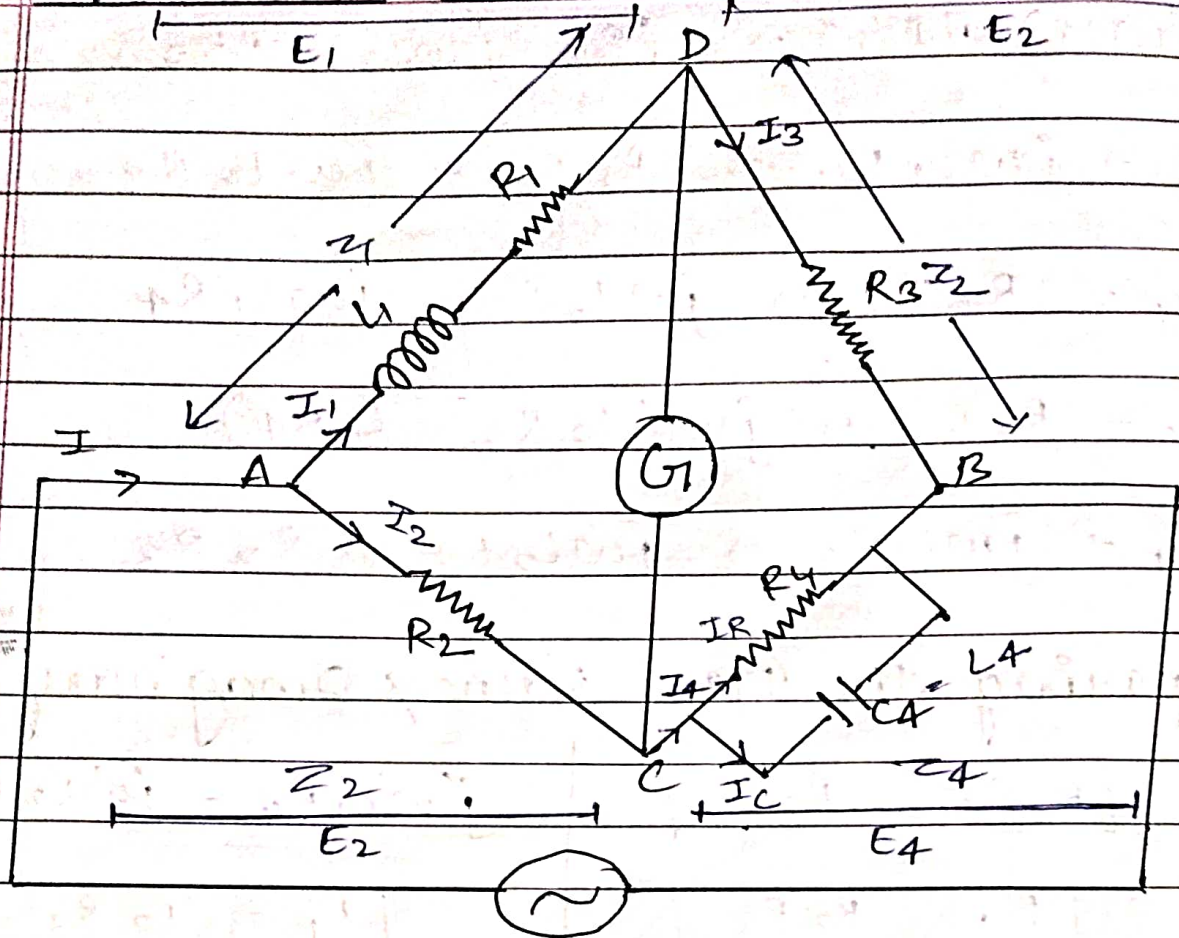
$$\boxed{R_1 \Rightarrow \frac{R_2 R_3}{R_4}}$$

$$\boxed{L_1 \Rightarrow \frac{L_2 R_3}{R_4}}$$

Phasor Diagram:-



MAXWELL'S INDUCTANCE - CAPACITANCE BRIDGE



- The Bridge Measures an Unknown Inductance in terms of a Known Capacitor.
- The use of standard arm offers the advantage of capacitance and Easy Shielding
- The Capacitor is almost a loss-less Component
- One arm has a resistance R_4 in parallel with L_4 and Hence, it is easier to write the balance equation using the admittance of arm 4 instead of the impedance.

$$Z_1 \Rightarrow R_1 + j\omega L_1$$

$$Z_2 \Rightarrow R_2$$

$$Z_3 \Rightarrow R_3$$

$$Z_4 \Rightarrow \frac{R_4 \times \frac{1}{j\omega L_4}}{R_4 + \frac{1}{j\omega L_4}} \Rightarrow \frac{R_4}{1 + j\omega L_4 R_4}$$

At balance Bridge -

$$Z_1 Z_4 \Rightarrow Z_2 Z_3$$

$$\Rightarrow (R_1 + j\omega L_1) \left(\frac{R_4}{1 + j\omega L_4 R_4} \right) \Rightarrow R_2 R_3$$

$$\Rightarrow \frac{R_1 R_4}{1 + j\omega L_4 R_4} + \frac{R_4 j\omega L_1}{1 + j\omega L_4 R_4} \Rightarrow R_2 R_3$$

$$\Rightarrow R_1 R_4 + R_4 j\omega L_1 \Rightarrow R_2 R_3 (1 + j\omega L_4 R_4)$$

$$\Rightarrow R_1 R_4 + R_4 j\omega L_1 \Rightarrow R_2 R_3 + j\omega L_4 R_4 R_2 R_3$$

on Comparing Real part and Imaginary Part-

$$\boxed{R_1 R_4 \Rightarrow R_2 R_3}$$

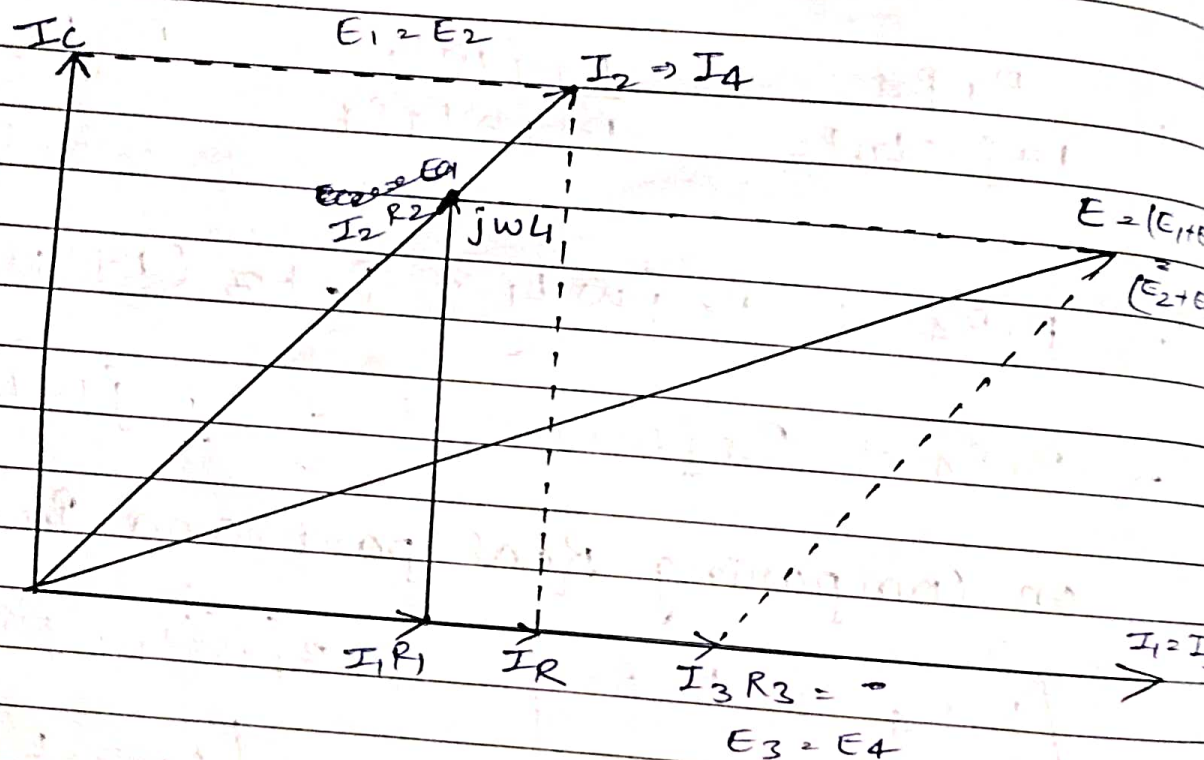
$$\cancel{R_4 j\omega L_1} = \cancel{j\omega L_4 R_4} R_2 R_3$$

$$\underline{R_1 \Rightarrow \frac{R_2 R_3}{R_4}}$$

$$\boxed{L_1 \Rightarrow L_4 R_2 R_3}$$

Phasor Diagram

Phasor Diagram



Quality factor = Q.F. = $\frac{W_4}{R_1}$

Q.F. = $\frac{W C_4 R_2 R_3}{R_2 \frac{R_3}{R_4}}$

This bridge is used for measurement of Inductance with Medium.

Q.F. = $1 < Q.F. < 10$ [Q.F. = $W C_4 R_4$]

Hay's Bridge

→ For L Measurement (AC Bridge)

$L_1 \rightarrow$ Unknown Resistance

$R_2, R_3, R_4 \rightarrow$ Known Resistance

$C_4 \rightarrow$ Standard Capacitor

$R_1 \rightarrow$ Unknown Resistance

$$Z_1 \rightarrow R_1 + j\omega L_1$$

$$Z_2 = R_2$$

$$Z_3 = R_3$$

$$Z_4 \rightarrow R_4 + \frac{1}{j\omega C_4}$$

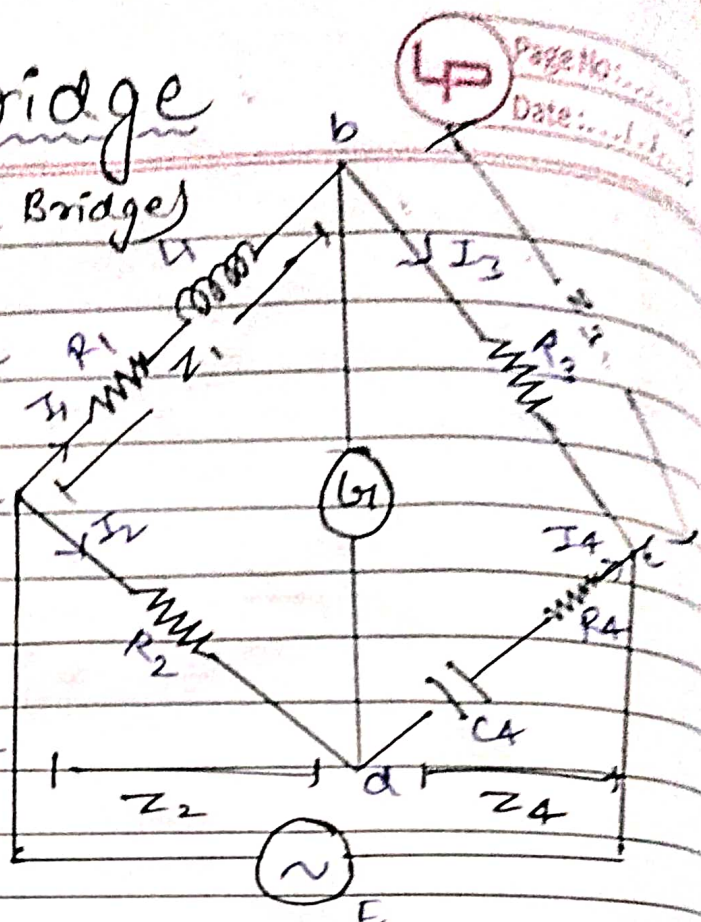
At Balanced Condition -

$$Z_1 Z_4 = Z_2 Z_3$$

$$\rightarrow (R_1 + j\omega L_1) \left(R_4 + \frac{1}{j\omega C_4} \right) \rightarrow R_2 R_3$$

$$\rightarrow R_1 R_4 + \frac{R_1}{j\omega C_4} + R_4 j\omega L_1 + \frac{j\omega L_1}{j\omega C_4} \rightarrow R_2 R_3$$

$$\rightarrow R_1 R_4 + \frac{R_1}{j\omega C_4} + R_4 j\omega L_1 + \frac{L_1}{C_4} \rightarrow R_2 R_3$$



Equating Real Part -

$$\cancel{R_1 R_4} + \frac{L_1}{\cancel{C_4}} = \cancel{R_2 R_3}$$

$$\rightarrow R_1 R_4 j\omega C_4 + R_1 + j\omega L_1 R_4 j\omega C_4 + j\omega L_1$$

$$\rightarrow R_2 R_3 j\omega C_4$$

$$\rightarrow R_1 R_4 j\omega C_4 + R_1 - \omega^2 L_1 R_4 C_4 + j\omega L_1 = R_2 R_3 C_4 j\omega$$

On Equating Real & Imaginary Part -

$$R_1 \rightarrow \omega^2 L_1 R_4 C_4$$

$$L_1 \rightarrow \frac{R_1}{\omega^2 R_4 C_4}$$

$$\therefore R_1 R_4 j\omega C_4 + j\omega L_1 \Rightarrow j\omega R_2 R_3 C_4$$

$$\Rightarrow \cancel{j\omega} (R_1 R_4 C_4 + L_1) \Rightarrow \cancel{j\omega} R_2 R_3 C_4$$

$$\rightarrow R_1 R_4 C_4 + L_1 \Rightarrow R_2 R_3 C_4$$

$$\text{put the value of } L_1 \Rightarrow \frac{R_1}{\omega^2 R_4 C_4}$$

$$\therefore \rightarrow R_1 R_4 C_4 + \frac{R_1}{\omega^2 R_4 C_4} \Rightarrow R_2 R_3 C_4$$

$$\Rightarrow \omega^2 R_1 R_4^2 C_4^2 + R_1 \Rightarrow \omega^2 R_2 R_3 R_4 C_4^2$$

$$R_1 (1 + \omega^2 R_4^2 C_4^2) \Rightarrow \omega^2 R_2 R_3 R_4 C_4^2$$

$$R_1 \Rightarrow \frac{\omega^2 R_2 R_3 R_4 C_4^2}{1 + \omega^2 R_4^2 C_4^2}$$

Phasor Diagram

